Ex No-2 Study of Network Cables.

AIM:-

To understand different types of network cables.

a) Different types of cables used in networking are

1) Unshielded twisted pair (UTP) cable
2) Shielded twisted pair (STP) cable
3) Coaxial cable.
4) Fibre optical cable

Cable Type	Category	Maximum data transmi	Advantages/ Disadvantages	Appl/ use	Image
UTP.	category 3	10bps	→ Advantages. cheap, easy to install. → Disadvantages More prone to ele-tromagnetic noise & interface.	10 base T - Ethernet	
	category 5	upto 100mbps		Fast ethernet Original ethernet	
	category 5e.	1Gbps.			
STP	category 6, 6e	10Gbps	Advantages → Shielded, faster than UTP Disadvantages → Expensive & Greater installation effort	Gigabit Ethernet, 10G (SSM) used in data centres	
	category 7	10Gbps			
Coaxial Cable.	RG-6, RG-59 RG-4.	10-100Mbps	Advantages → High bandwidth → Immune to interface Disadvantages → Limited distance cost.	Television Network	
fibre optics cable.	single mode, Multi Mode.	100mbps	Advantages →High speed, bandwidth, Security Disadvantages →Costly, skilled installers req	Maximum dist of fibre optics cells is around 100cm.	

(b) Make your own ethernet cross-over cable / straight cable

Tools required

→ Ethernet cabling CAT5e is required for gigabit support, but CAT5 cabling works as well just over short distance

→ A chrimping tool, Two RJ45 plugs, Two plug shields.

Step 1:-

To start construction of a device, begin by threading shields onto the cable.

Step 2:-

Next, strip approximately 1.5 cm of cable shielding from both the ends. The chrimping tool has a round area to compelete the task.

Step 3:-

After, you will need to untangle the wires, these should be four 'twisted pairs' referencing back to the sheet, arrange them from top to bottom. One End, should be in arrangement A and the other in B.

Step 4:-

Once the order is correct, bunch them togethere in a line and if there are any that stick out farther then others, snip them back to create an even level. The difficult aspect is placing these into the RJ 45 plug without messing up the order To do so, hold the plug with the clip side facing away from you & have the goldpins facing towards you as shown.

Steps:-

Next push the cable right in. The notch at the end of the plug need to be just over the cable shielding and if it isn't that means you stripped off too much. shielding simply snip the cable back a little more.

Step 6:-

After the wires are securely sitting inside the plug insert it into the crimping tool & push down

Step 7:-

Lastly repeat for the other end using diagram B (to make crossover cables) / using diagram A / to make straight through cables)

Student observation.

→ Straight cables connect dissimilar devices by keeping the pin order identical on both ends.

→ Crossover cables connect similar devices by swapping the transmit & receive wire pairs internally.

→ A crossover cable is used to connect two PCs directly to each other.

→ A straight cable is used to connect a router or a switch to your PC

RESULT:-

Thus different types of cable networks are learned successfully.